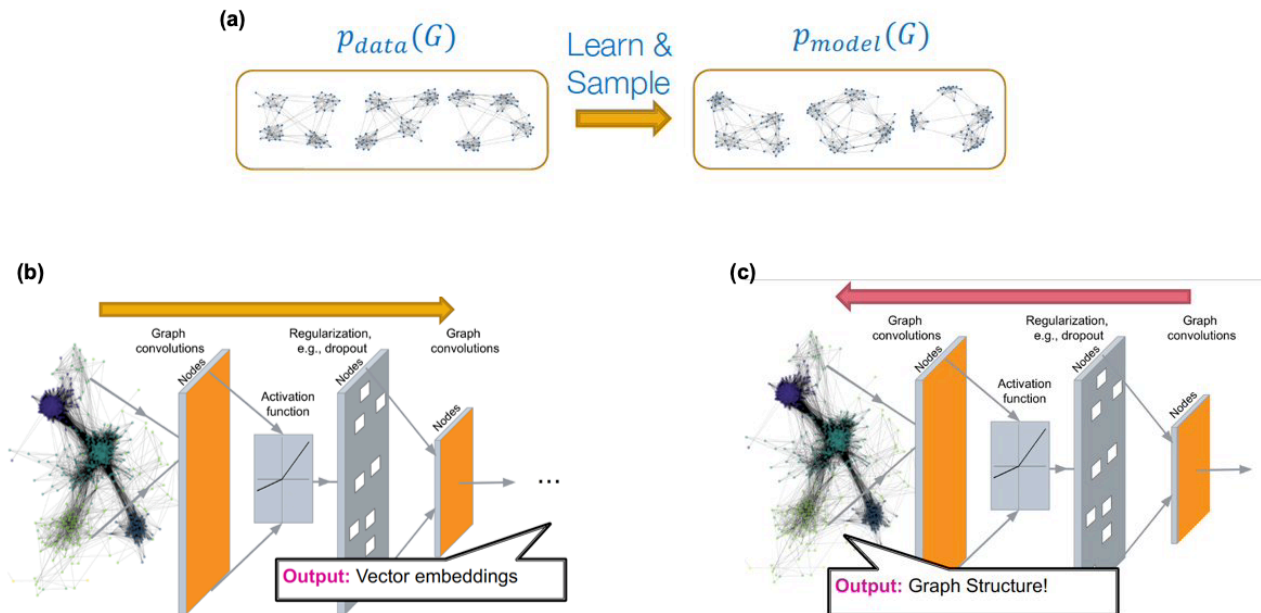


L14: Advanced Topics: Deep Generative Models for Graphs



Source: "Machine Learning with Graphs" – Jure leskovec <http://web.stanford.edu/class/cs224w>

We have already studied several network generation models in this course (mostly in Lesson-12 but also in earlier lessons). For instance, we have seen models with only one or two parameters (such as the Erdős-Rényi model or the preferential attachment model) as well as models with more parameters that can create networks with modular or hierarchical structure (such as stochastic block models). All of these models however are based on **explicit**, or **prescribed**, assumptions about the desired structure of the network. For example, we choose to use the preferential attachment model if we want to create a network with a power-law degree distribution of a certain exponent.

What if we want to generate networks that have similar structure with one (or more) given real networks -- but we do not have an explicit structural characterization of those networks? For example, given a portion of the Facebook friendship graph, we may be asked to create several synthetic networks that have the same structural properties with the Facebook graph -- even though we may not be in a position to list (or even know!) all those structural properties.

As shown in part-(a) of the above Figure, the goal in that case is to use one or more given graphs (the data) to learn a high-dimensional joint distribution (the model) that gives the probability that any two nodes are connected given the connectivity between all other nodes in a graph. We can then use that model to create many synthetic networks, of arbitrary size, with the expectation that all those networks share the same structural properties with the given data.

The main challenges in developing graph generative models are:

- the generative model should be able to generate graphs of arbitrary size and density. In practice this is difficult if the given data only refer to networks of a given size and density (e.g., it is hard to generate realistic dense networks if all the data refer to sparse networks).
- the generative model needs to consider the issue of "graph isomorphism", meaning that some networks may appear different (e.g., we may have ordered/labeled the nodes in different ways) even though the networks are identical (or almost identical) in terms of structure.
- the generative model needs to learn all non-trivial structural properties of the given graphs, such as degree correlations, community structure, hierarchy, and many others, without being explicitly trained to capture any of these properties.

This is an active research area that spans both network science and machine learning. Most of the state-of-the-art "generative graph models" rely on deep neural networks, as shown in Figures (b) and (c) above. The deep learning architectures that have

been adapted in this context are: variational auto-encoders, deep Q-networks (reinforcement learning), generative adversarial networks (GANs), and generative recursive neural networks (RNNs). We will not discuss these approaches in more depth here because they require prior knowledge in deep learning.

If you are interested to learn more about this topic, we recommend the 2020 article: [**"A systematic survey on deep generative models for graph generation"**](https://arxiv.org/abs/2007.06686)  [\(https://arxiv.org/abs/2007.06686\)](https://arxiv.org/abs/2007.06686) by X.Guo and L.Zhao.

One application in which graph generative models has attracted large interest is in computational chemistry (molecule design) and drug synthesis. Discovering a new molecule that is similar to existing molecules (e.g. a given set of antibiotics) is a very expensive and time-consuming process, especially if every candidate molecule is to be chemically synthesized and tested at the lab. Network science and deep learning techniques have been used to propose good molecular candidates (graphs between chemical elements) that have a high "score" in terms of their similarity to existing molecules.

The additional challenge, however, is that these candidate networks need to represent valid molecules. For instance, they need to satisfy the valency of each atom in the molecule so that the resulting chemical is stable. Another challenge is that the candidate networks should be, not just similar, but better than the given molecules in terms of various properties (e.g., a new antibiotic should be effective to bacteria that are already resistant to the existing antibiotics that are used as data in the corresponding generative model).

A recent survey of methods in this area is [**"Deep learning for molecular design"**](https://pubs.rsc.org/en/content/articlelanding/2019/me/c9me00039a/unauth#!divAbstract)  [\(https://pubs.rsc.org/en/content/articlelanding/2019/me/c9me00039a/unauth#!divAbstract\)](https://pubs.rsc.org/en/content/articlelanding/2019/me/c9me00039a/unauth#!divAbstract) by D.Elton et al.